

Name.....

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- MARCH 2023

SIXTH SEMESTER – (HONOURS)

CST 394 NETWORK SECURITY

Time: 2 Hours

Max.Marks: 50

PART A

Answer all Questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	Distinguish between signature-based and anomaly-based intrusion detection techniques.	3	K3	CO1&K3
2	Explain about Digital Signature Standard (DSS)	3	K2	CO1&K3
3	Describe the management functions of a PKI.	3	K2	CO2&K2
4	What are the benefits of IPsec	3	K1	CO2&K2
5	Explain structure of a PEM Message with an example.	3	K2	CO3&K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Module 1: Qn. No 6,7 and 8 – Answer any 2

Module 2: Qn. No 9,10 and 11 – Answer any 2

Module 3: Qn. No 12 is mandatory

Qn. No.	Questions	Marks	K Level	CO & KL
6	What are the requirements and challenges of network security.	7	K2	CO1&K3
7	Explain the network security model with the help of a neat diagram.	7	K2	CO1&K3
8	Differentiate between Worms and Viruses.	7	K3	CO1&K3
9	What is Kerberos? Explain how it provides authenticated service.	7	K2	CO2&K2
10	What is the Authentication Header in IPsec.	7	K1	CO2&K2
11	Draw the ESP packet format and explain it.	7	K1	CO2&K2
12	What are the security services for Electronic Mail.	7	K1	CO3&K3

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Time: 2 Hours

Max. Marks: 50

PART AAnswer **ALL** questions (3x5=15 marks)

Qn No.	Questions	Marks	K level	CO & KL
1	Interpret the differences between authentication and message integrity.	3	K2	CO3 & K3
2	Explain the role of S/MIME in security.	3	K2	CO3 & K3
3	Discuss briefly the threats in web security.	3	K2	CO4 & K2
4	Describe the architecture of secure socket layer(SSL)	3	K2	CO4 & K2
5	Define WAP.	3	K1	CO5 & K2

PART BAnswer **Five** questions (7x5=35 marks)

Qn No.	Questions	Marks	K level	CO & KL
6	Explain the security standards used in network communication	7	K2	CO2 & K2
7	Demonstrate the mechanisms in email security services	7	K3	CO3 & K3
8	Explain the fundamental concepts of wireless network security and firewalls	7	K2	CO5 & K2
9	Describe the protocols used to provide web security	7	K2	CO4 & K2
10	Demonstrate the working of PEM.	7	K3	CO3 & K3
11	Illustrate the application of PGP in email security.	7	K3	CO3 & K3

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech (Hons.) Degree Examination June 2023 (2020 Admission)

Course Code: CST394**Course Name: NETWORK SECURITY**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|-----|---|-----|
| ✓1 | List and define the three main objectives of computer security. | (3) |
| 2 | What are the different phases in the life time of a virus? | (3) |
| 3 | Define Perfect forward secrecy and write its applications. | (3) |
| 4 | Write any three limitations of Kerberos Version 4. | (3) |
| ✓5 | Explain the steps in the working of Privacy Enhanced Mail (PEM). | (3) |
| ✓6 | Define non repudiation with an example. | (3) |
| ✓7 | Describe the privacy concerns that occur while browsing the web. | (3) |
| ✓8 | How an HTTP client can perform the closing of a connection. | (3) |
| ✓9 | Write two security measures taken by Wi-Fi Protected Access (WPA). | (3) |
| ✓10 | Define the terms Access Point, Basic Service Set, Extended Service Set. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) User P wants to send a message "wait till further orders" to user Q. Based on Network security model, apply any cryptographic algorithm to plain text and write the transformations of the message at various stages and explain the role of each component of Network security model with diagram. (7)
- b) Let prime number $q=19$, primitive root $\alpha=10$ and message with hash value $m=14$. Generate digital signature for user A using Elgamal Digital Signature Scheme and verify the same by user B. (7)

OR

- 12 a) Discuss different types of Intrusion Detection Systems with strengths and drawbacks. (7)
- b) Write Schnorr digital signature scheme. (7)

Module II

- 13 a) Define Public Key Infrastructure (PKI), draw the architectural model of PKI and illustrate the functions of key elements. (7)

- b) Draw the structure of Encapsulating security payload (ESP) packet and explain the functions of each component. (7)

OR

- 14 a) Design a system based on the principle of Kerberos, which performs authentication of the user and provide a service. Draw the process diagram and illustrate the working. (7)
- b) Write the features of Internet Key Exchange (IKE) key determination algorithm. (7)

Module III

- 15 a) Illustrate the four classifications of email security threats. Enlist the standardised protocols, which can be used to counter these threats. (7)
- b) Describe the message content types used by Secure /Multipurpose Internet Mail Extension (S/MIME). List the approved cryptographic algorithms used in S/MIME with its functions. (7)

OR

- 16 a) Illustrate the services provided by PGP and its working with a suitable diagram. (8)
- b) Differentiate between S/MIME and OpenPGP. (6)

Module IV

- 17 a) Draw the Secure Socket Layer (SSL) architecture and describe the functions of the components. (7)
- b) Explain the attacks on Secure Socket Layer (SSL) / Transport Layer Security (TLS). (7)

OR

- 18 a) Illustrate the sequence of events in the Secure Shell (SSH) transport layer protocol packet exchange with a diagram. (8)
- b) Define HTTPS. Which are the elements of communication, get encrypted when HTTPS is used. (6)

Module V

- 19 a) Explain the IEEE 802.11i phases of operation with a diagram. (7)
- b) Explain the two authentication methods used in Wired Equivalent Privacy (WEP). (7)

OR

- 20 a) Why are firewalls needed? Compare the features of packet filters and circuit level firewalls. (8)
- b) Describe the three transaction classes provided by Wireless Transaction Protocol. (6)
